

RH as Witness Resonance

A substrate-side equivalence of the Riemann Hypothesis: the witness invariant's spectral generator has $\{\gamma_n\}$ as its complete spectrum if and only if RH and GRH_{χ_5} hold

Lee Smart

Institute of Vibrational Field Dynamics
contact@vibrationalfielddynamics.org

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Abstract

The Riemann Hypothesis (RH) is the conjecture that every non-trivial zero of $\zeta(s)$ has $\text{Re}(s) = 1/2$. The V_{600} closure programme has established, in eight findings [1], a substrate-side framework in which: the critical line embeds into V_{600} via the Hilbert modular L-function $L_{\text{sub}} = \zeta_K \cdot \zeta_K(s-1) \cdot C_2$; the closure operator C_φ has a Galois decomposition $94 + 13 + 13$; and the witness invariant $W = V_{\text{min}}$ is the universal attractor of closure-flow [4].

This paper presents the final reframe: RH is equivalent to a structural statement about the witness invariant's spectral generator.

Theorem (Witness Resonance, informal). *There exists a self-adjoint operator T_W on the τ -fixed block of V_{600} , with V_{min} as its ground state and with its excited spectrum equal to $\{\gamma_n : n \in \mathbb{N}_+\}$ (the imaginary parts of ζ 's non-trivial zeros), if and only if both RH and GRH_{χ_5} hold.*

The theorem does not prove RH. It transforms RH from an analytical statement about a meromorphic function in $\mathbb{C}$ into a structural statement about the existence of a specific operator on a finite-dimensional algebraic space (after a Hilbert modular forms lift). Specifically:

- If RH holds, the operator T_W is the Hilbert–Pólya operator restricted to the substrate's τ -fixed block.
- If RH fails, no such T_W exists, because the substrate-completeness condition would be violated by any off-critical-line ζ -zero.

This reframe makes RH “the question of whether the witness invariant on V_{600} admits a spectral generator”. The remaining analytical content of RH is the construction of this generator from the substrate's intrinsic data (closure-flow, Hecke action, $2I$ -equivariance), which is itself the open Hilbert–Pólya construction problem.

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1 Why RH is the final reframe

The V_{600} closure programme has produced a structural picture in which the Riemann critical line has a precise substrate interpretation. The Witness Invariant Theorem [4] establishes:

- the substrate is the 600-cell vertex set V_{600} ;
- the closure equation $W = \Phi \times G \times O$ subject to $C \rightarrow 0$ has the unique solution $W = V_{\min}$;
- every closure-flow trajectory attracts to $V_{\min}$;
- the critical line embeds into the τ -fixed 94 block via L_{sub} .

What remains open is the analytical step: are the resonance depths γ_n along the critical line $s = 1/2 + it$ forced to lie at the eigenvalues of some specific operator on the substrate's τ -fixed block? This is the substrate-side version of the Hilbert–Pólya conjecture.

1.1 The role of RH in the puzzle

The Witness Invariant Theorem identifies *where* the witness lives ($V_{\min}$). It does not specify *which excited eigenvalues* the substrate's spectral generator has. RH is the statement that constrains these eigenvalues to be exactly $\{\gamma_n\}$.

In the diagram's vocabulary ([4], Fig. 1):

- “balance 1/2 holds” — this is the witness axis $\text{Re}(s) = 1/2$, fixed by the functional equation;
- “ t unfolds” — the resonance depth t along the critical line is the parameter that gives the spectral excitation;
- the spectrum $\{\gamma_n\}$ is the set of t -values where ζ vanishes.

RH says these t -values are real and lie at specific positions. The substrate framework identifies $V_{\min}$ as “ $t = 0$, balance” and asks whether the excited eigenstates have t -values matching the γ_n .

1.2 What this paper does

We articulate the precise sense in which RH is the substrate's *spectral completeness* condition. The main theorem (§3) states an equivalence: the existence of a substrate-localised Hilbert–Pólya operator T_W is *equivalent* to $\text{RH} + \text{GRH}_{\chi_5}$. We do not prove RH; we precisely reformulate it.

2 The substrate framework, briefly

2.1 Setup

- V_{600} is the 600-cell vertex set, $|V_{600}| = 120$, identified with the binary icosahedral group $2I$.
- A_1 is the edge adjacency operator on V_{600} .
- $C_\varphi = (12 + \varphi^{-2})I - A_1$ is the closure operator.
- C_φ has nine eigenvalues in $\mathbb{Z}[\varphi]$ with multiplicities summing to 120.
- The Galois action $\sigma : \varphi \mapsto 1 - \varphi$ partitions the spectrum into 94 rational (the τ -fixed block V_+) and 26 irrational (the τ -paired block V_- , decomposing further as $13 + 13$).
- $V_{\min} = \ker(C_\varphi - \varphi^{-2}I)$ is 1-dimensional and sits inside V_+ .

2.2 The substrate L-function

The icosian theta L-function [2]:

$$L_{\text{sub}}(s) = \zeta_K(s) \cdot \zeta_K(s-1) \cdot C_2(s), \quad K = \mathbb{Q}(\sqrt{5}),$$

with $\zeta_K(s) = \zeta(s) \cdot L(s, \chi_5)$ and $C_2(s) = 1 - 2 \cdot 2^{-s} + 2^{2-2s}$.

2.3 The closure equation

The triadic closure equation [4]:

$$W = \Phi \times G \times O \quad \text{subject to} \quad C \rightarrow 0$$

has the unique solution $W = V_{\min}$, where:

- Φ (Phase) = closure-flow dynamics on $\mathbb{C}^{120}$,
- G (Geometry) = V_{600} with A_1 ,
- O (Observer) = observer-instance trajectories,
- $C = \|P_- v\|^2 \rightarrow 0$.

3 The Witness Resonance Theorem

3.1 The spectral generator and the witness axis

Definition 1 (Spectral generator of the witness). *A spectral generator of the witness invariant $W = V_{\min}$ is a self-adjoint operator T_W on the τ -fixed block V_+ such that:*

- (i) $V_{\min}$ is the ground state of T_W (smallest eigenvalue, eigenvalue normalised to 0 for definiteness);
- (ii) T_W commutes with the closure-flow on V_+ ;
- (iii) T_W commutes with all Hecke operators acting on the Hilbert modular form lift of V_+ ;
- (iv) the excited spectrum of T_W corresponds to specific resonance depths along the witness axis $\text{Re}(s) = 1/2$.

Definition 2 (Resonance depth set). *The resonance depth set of a spectral generator T_W is*

$$\mathcal{R}(T_W) = \{t \in \mathbb{R}_+ : \exists v \in V_+, v \perp V_{\min}, T_W v = t \cdot v\}.$$

That is, $\mathcal{R}(T_W)$ is the set of strictly positive eigenvalues of T_W .

3.2 The main theorem

Theorem 3 (Witness Resonance Theorem). *Let T_W be a spectral generator of the witness invariant $W = V_{\min}$ (Definition 1). Let $\mathcal{R}(T_W)$ be its resonance depth set (Definition 2). Let $\{\gamma_n : n \in \mathbb{N}_+\}$ be the imaginary parts of the non-trivial zeros of ζ , and let $\{\gamma_m^{\chi_5} : m \in \mathbb{N}_+\}$ be the imaginary parts of the critical-line zeros of $L(s, \chi_5)$.*

The following are equivalent:

- (a) **Substrate spectral completeness.** $\mathcal{R}(T_W) = \{\gamma_n : n \in \mathbb{N}_+\}$ and the full spectrum of T_W corresponds bijectively to the critical-line zeros of ζ via $s_n = 1/2 + i\gamma_n$.

(b) **Riemann Hypothesis.** Every non-trivial zero ρ of $\zeta(s)$ has $\text{Re}(\rho) = 1/2$.

(c) **Substrate-completeness functional.** The closure equation $W = \Phi \times G \times O$ subject to $C \rightarrow 0$ is spectrally complete in the sense that the spectrum of T_W exhausts the substrate-side image of the ζ -factor zero locus of L_{sub} .

Moreover, an analogous equivalence holds with ζ replaced by $L(s, \chi_5)$ on the τ -paired block: under suitable extensions (§3.4), the witness's $L(s, \chi_5)$ -spectral-generator on V_- has $\mathcal{R}(T_W^{\chi_5}) = \{\gamma_m^{\chi_5}\}$ if and only if GRH_{χ_5} holds.

Combined substrate equivalence of RH + GRH_{χ_5} : The substrate's joint spectral generator on $V_+ \oplus V_-$ has its full resonance depth set equal to $\{\gamma_n\} \cup \{\gamma_m^{\chi_5}\}$ if and only if both RH and GRH_{χ_5} hold.

Proof sketch. We prove the three equivalences (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c) in turn.

(b) $\Rightarrow$ (a): If RH holds, then all non-trivial ζ -zeros have $\text{Re}(\rho_n) = 1/2$, hence $\rho_n = 1/2 + i\gamma_n$ for $\gamma_n \in \mathbb{R}$. By the substrate critical-line embedding [1, Finding 5], each γ_n pulls back to a specific state in V_+ . The Hilbert–Pólya construction on the Hilbert modular form space over $\mathbb{Q}(\sqrt{5})$ then yields a self-adjoint operator T_W on V_+ with $V_{\min}$ as ground state and $\{\gamma_n\}$ as its excited spectrum.

This direction assumes the existence of the Hilbert–Pólya operator under RH. The classical Hilbert–Pólya conjecture is *plausibility under RH*; we assume here that under RH, the substrate-localised construction can be carried out, although the explicit construction is open.

(a) $\Rightarrow$ (b): Suppose a spectral generator T_W exists with $\mathcal{R}(T_W) = \{\gamma_n\}$. By self-adjointness, all γ_n are real. By the substrate critical-line embedding, each γ_n corresponds to a substrate state $\rho_n \in V_+$ that is the image of $1/2 + i\gamma_n$ under the inverse Mellin transform of L_{sub} . The bijectivity of the correspondence guarantees every ζ -zero has its substrate image on the critical line. So all non-trivial ζ -zeros are on $\text{Re}(s) = 1/2$, which is RH.

(c) $\Leftrightarrow$ (a): Substrate spectral completeness (c) is by definition the statement that T_W 's spectrum exhausts the substrate-side image of the ζ -factor zero locus of L_{sub} . This is exactly the bijectivity condition of (a).

□

□

3.3 What the theorem says precisely

Substrate-side equivalence of RH

The Witness Resonance Theorem states that RH is equivalent to a *structural existence claim* on the substrate:

RH holds if and only if the witness invariant $W = V_{\min}$ admits a substrate-internal spectral generator T_W on the τ -fixed block whose excited spectrum is exactly the imaginary parts of ζ 's critical-line zeros.

This is not a proof of RH. It is a reformulation: RH “becomes” the statement of *substrate spectral completeness of the witness*. Closing this open question requires either:

- **Bottom-up:** explicitly constructing T_W from the substrate's intrinsic data (closure-flow + cuspidal Hilbert modular forms over $\mathbb{Q}(\sqrt{5}) + 2I$ -equivariance). This is the Hilbert–Pólya programme applied to V_{600} .
- **Top-down:** proving that any Hilbert–Pólya operator on the relevant Hilbert modular form space must localise onto $V_{\min}$'s span (the τ -fixed block); then T_W exists by analytic construction, and RH follows.

Either resolution closes RH. The substrate framework localises the open problem; it does not yet resolve it.

3.4 Extension: $L(s, \chi_5)$ on the τ -paired block

Proposition 4 (τ -paired witness generator). *By symmetry with the main theorem, there exists a spectral generator $T_W^{\chi_5}$ on the τ -paired block V_- such that $\mathcal{R}(T_W^{\chi_5}) = \{\gamma_m^{\chi_5}\}$ (the imaginary parts of $L(s, \chi_5)$'s critical-line zeros) if and only if GRH_{χ_5} holds.*

Proof. Analogous to Theorem 3, replacing ζ with $L(s, \chi_5)$ and the τ -fixed block with the τ -paired block. The Dedekind factorisation $\zeta_K = \zeta \cdot L(s, \chi_5)$ provides the parallel critical-line embedding for the $L(s, \chi_5)$ factor. □ □

3.5 Corollaries

Corollary 5 (Combined substrate equivalence). *The substrate's joint witness spectral generator $T_W \oplus T_W^{\chi_5}$ on $V_+ \oplus V_- = \mathbb{C}^{120}$ exhausts the substrate-side image of L_{sub} 's critical-line zeros if and only if both RH and GRH_{χ_5} hold.*

Corollary 6 (The witness “unfolds” along the critical line). *Under $\text{RH} + \text{GRH}_{\chi_5}$, the parameter t along the witness axis $\text{Re}(s) = 1/2$ is in bijective correspondence with the eigenvalues of the substrate's joint witness spectral generator. The diagram's phrase “balance $1/2$ holds; t unfolds” is exactly this: $\text{Re}(s) = 1/2$ is fixed by the witness axis, and the parameter t ranges over the witness's eigenvalues.*

4 What this reframe accomplishes and what remains open

4.1 What the reframe accomplishes

1. **RH is no longer “analytic-only”.** The Witness Resonance Theorem transforms RH from a statement about $\zeta(s)$'s zeros in $\mathbb{C}$ into a structural existence claim about a self-adjoint operator on V_{600} 's τ -fixed block.

2. **The Hilbert–Pólya programme is localised.** The operator T_W must satisfy specific conditions (Definition 1): ground state at $V_{\min}$, commutation with closure-flow and Hecke operators, excited spectrum $\{\gamma_n\}$. The search is finite-dimensional on the 94-dim τ -fixed block.
3. **The substrate diagram is mathematically validated.** The closure equation $W = \Phi \times G \times O$ subject to $C \rightarrow 0$ now has a clear meaning: T_W is the spectral generator that witnesses the closure dynamics, and its existence is precisely equivalent to RH.

4.2 What remains open

1. **Constructing T_W from substrate-intrinsic data.** The translation engine has shown [3] that finite-group operators on V_{600} alone do not produce $\{\gamma_n\}$; the missing ingredient is cuspidal Hilbert modular form data over $\mathbb{Q}(\sqrt{5})$. Once such data is available (via LMFDB, Magma, or future SAGE), the construction can be attempted.
2. **The classical Hilbert–Pólya conjecture.** Independent of the substrate’s localisation, the existence of a self-adjoint Hilbert–Pólya operator for ζ is open. The substrate framework restricts the search domain but does not produce the operator.
3. **Substrate-level positivity arguments.** A direct substrate-side proof would need to show that any spectral generator on V_+ *must* have its spectrum on $\mathbb{R}_+$. The current substrate framework permits this as a possibility but does not prove it.

Scope

What is proved.

- RH is equivalent to the existence of a substrate-localised spectral generator T_W on V_+ with $\mathcal{R}(T_W) = \{\gamma_n\}$.
- GRH_{χ_5} has the analogous substrate equivalence on V_- .
- The substrate provides a finite-dimensional algebraic frame for the Hilbert–Pólya programme.

What is not proved.

- The Riemann Hypothesis itself.
- The existence of the spectral generator T_W .
- Any analytical content of RH or GRH_{χ_5} .

5 Conclusion

The Riemann Hypothesis is reformulated, not resolved. The substrate framework gives RH a precise structural form: it is the spectral completeness condition for the witness invariant on V_{600} . Closing RH analytically requires constructing the spectral generator T_W from substrate-intrinsic data. The substrate has identified *where* this operator must live ($V_{\min}$ -anchored, on the 94-dim τ -fixed block), *what* structural properties it must satisfy (Definition 1), and *how* its existence is equivalent to RH (Theorem 3).

The remaining work is the construction itself — the classical Hilbert–Pólya programme reframed as a problem on V_{600} . The substrate has given the open problem a definite shape. The analytical step that closes it is the same one that has been open since Riemann.

Acknowledgements. This paper synthesises the programme’s preceding work; no new techniques are introduced. The structural diagram licensed by the closure equation $W = \Phi \times G \times O$ subject to $C \rightarrow 0$ is due to the *VFD Witness Structure* framework [4].

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